

1a)

$$\begin{aligned} -3p^2 + 6p - 2 &= 0 \\ p &= \frac{-6 \pm \sqrt{6^2 - 4 \cdot (-3) \cdot (-2)}}{2 \cdot (-3)} = \frac{-6 \pm \sqrt{12}}{-6} = 1 \mp \frac{\sqrt{3}}{3} \\ p_1 &\approx 0.423, \quad p_2 \approx 1.577 \end{aligned}$$

$a < 0$, joten $r(p) > 0$ nollakohtien välissä: hintaväli $[0.423, 1.577]$.

1b)

$$\begin{aligned} r'(p) &= -6p + 6 = 0 \Rightarrow p = 1 \\ r(1) &= -3 + 6 - 2 = 1 \end{aligned}$$

Suurin voitto on 1.

2a)

$$\begin{aligned} 2015: \quad p(7) &= 54.99 \cdot 1.0249^7 \approx 65.32 \\ 2025: \quad p(17) &= 54.99 \cdot 1.0249^{17} \approx 83.54 \end{aligned}$$

3a)

$$\begin{aligned} 3 \cdot \left(\frac{1}{2}\right)^{t/1600} &= 0.67 \cdot 3 \\ \frac{t}{1600} \ln \frac{1}{2} &= \ln 0.67 \\ t &= \frac{1600 \ln 0.67}{\ln 0.5} \approx 924.4 \\ 1954 + 924.4 &\approx 2878.4 \end{aligned}$$

Vuonna 2878.

3b)

$$\begin{aligned} 54.99 \cdot 1.0249^t &= 100 \\ t \ln 1.0249 &= \ln \frac{100}{54.99} \\ t &= \frac{\ln(100/54.99)}{\ln 1.0249} \approx 24.3 \\ 2008 + 24.3 &\approx 2032.3 \end{aligned}$$

Vuonna 2032.

4a) $76 - 71.7 = 4.3$ prosenttiyksikköä.

4b) $\frac{76 - 71.7}{71.7} \approx 0.060 = 6.0 \%$

4c) $\frac{71.5 - 71.7}{71.7} \approx -0.0028$, eli laskenut 0.28 %.